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AGENTIC THREAT HUNTING

HUNT WITH MEMORY. HUNT WITH AGENTS. · DEATHCON 2026

AGENTIC THREAT HUNTING FRAMEWORK — OPEN SOURCE, YOURS TO KEEP.

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Founder of **THOR Collective**.

DEATHcon 2026

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BEFORE YOU PRESS PLAY — PAUSE POINT 0

```
python3 -m venv .venv && source .venv/bin/activate     # Windows: .venv\Scripts\activate
pip install 'agentic-threat-hunting-framework[all]'
cp providers/<your-provider>.env.example .env          # edit: paste your API key
python preflight/validate.py                           # all green = you're ready
```

- **Splunk:** the URL + login DEATHcon gave you · read-only · time picker **All time** · `index=thrunt`
- **MCP client:** point it at `athf-mcp` now, not in Module 4 — `providers/mcp.md`
- Everything: `github.com/Nebulock-Inc/agentic-hunting-workshop-deathcon-2026` · new to hunting? `PRE_READ.md`**, 10 min**

TIME RANGE: ALL TIME. EVERY SEARCH. EVERY TIME.

THE FIVE LEVELS OF AGENTIC HUNTING

LEVEL	NAME	WHAT IT MEANS	ATHF
L0	**AD-HOC**	Hunt in a search bar. Close laptop, hunt is gone.	—
L1	**DOCUMENTED**	Write the hunt down so it survives you.	`hunt new`
L2	**SEARCHABLE**	Find the next hunt by reusing the last.	`similar`
L3	**GENERATIVE**	Agent reads intel, hands you hypotheses.	`research new`
L4	**AGENTIC**	Agent has tools; finds what you haven't covered.	mcp

THE MEMORY IS THE MOAT.

LOCK — HOW A HUNT IS SHAPED

- **L — Learn:** what triggered this hunt? Intel, an anomaly, a gap.
- **O — Observe:** the hypothesis and the query. What behavior would this actor leave?
- **C — Check:** run it, triage it, rule things out. Dead ends count.
- **K — Keep:** the finding, the SPL, the verdict. `athf similar` reads this later.

> A ruled-out hypothesis is still a finding.

TEN COMMANDS. THAT'S THE WHOLE SESSION.

COMMAND	WHEN
`athf init`	ONCE, MODULE 1. ENTER THROUGH THE PROMPTS.
`athf hunt new …`	EVERY HUNT. OR COPY A TEMPLATE.
`athf hunt list` · `validate`	END OF EVERY LAB
`athf similar` · `hunt coverage`	MODULE 2 ONWARD
`athf research new`	MODULE 3
MCP: *"what should I hunt next?"*	MODULE 4 ONWARD

ALL OF IT IS IN `COMMANDS/`. RUN THE FILE; DON'T RETYPE IT.

A HUNT IS JUST A FILE

- `athf hunt new --title … --technique … --behavior …` scaffolds a Markdown record in `hunts/`.
- **Or skip the flags:** copy `templates/H-0004.md` into `hunts/`, fill in **Check** and **Keep** in your editor.
- Either way, `athf hunt validate` tells you if it's complete. `athf similar` reads it the same.
- The typing is not the lesson. **The writing-it-down is.**

THE SHAPE OF THE SESSION

#	MODULE	VIDEO	LAB
1	MANUAL + STRUCTURE	11 min	25 min
2	SEARCH + SIMILARITY	9 min	20 min
3	RESEARCH AGENTS	8 min	15 min
4	AGENTIC MCP	9 min	15 min
5	THE PIVOT — NO HAND-HOLDING	6 min	15 min

PAUSE SLIDE IN, DEBRIEF OUT, EVERY LAB. WATCH THE GAP OPEN.

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MANUAL HUNTING & STRUCTURE

FEEL THE PAIN OF AD-HOC HUNTING. EARN THE CURE.

MEET THE VICTIM: SUDO SHIPPING. FLEET LOGISTICS, DEVS MOVE FAST. THE SOC DOESN'T KNOW YET.

WHY WE START BY HAND

- **This is hunting, not triage.** No alert fired. Assume compromise, go looking with a hypothesis.
- **The intrusion:** Sudo Shipping — devs install build tooling from a public registry, no gate.
- A backdoored patch of a popular CLI ran at install time, with developer privileges.
- **The binary is signed. The hash is clean in every feed. There is no CVE.**
- You cannot hash your way out of this. The only way in is behavior.

THE BEHAVIORAL MINDSET

- **Pyramid of Pain:** hashes and IPs are trivial to change; TTPs are expensive.
- CRON GOBLIN gives you nothing at the bottom — clean hash, CDN-looking domain.
- **Hunt the shape, not the string:** trusted-dev-process → unexpected shell → recon.
- A `node` post-install spawning `cmd.exe`, then `whoami /all` + `net group "Domain Admins"` — that's not a build step.

ABLE — THE HYPOTHESIS IS A SENTENCE

Actor	CRON GOBLIN
Behavior	POST-INSTALL SCRIPT SPAWNS A SHELL
Location	DEVELOPER WORKSTATIONS
Evidence	`NODE` → `CMD.EXE` PARENT/CHILD IN EDR PROCESS EVENTS

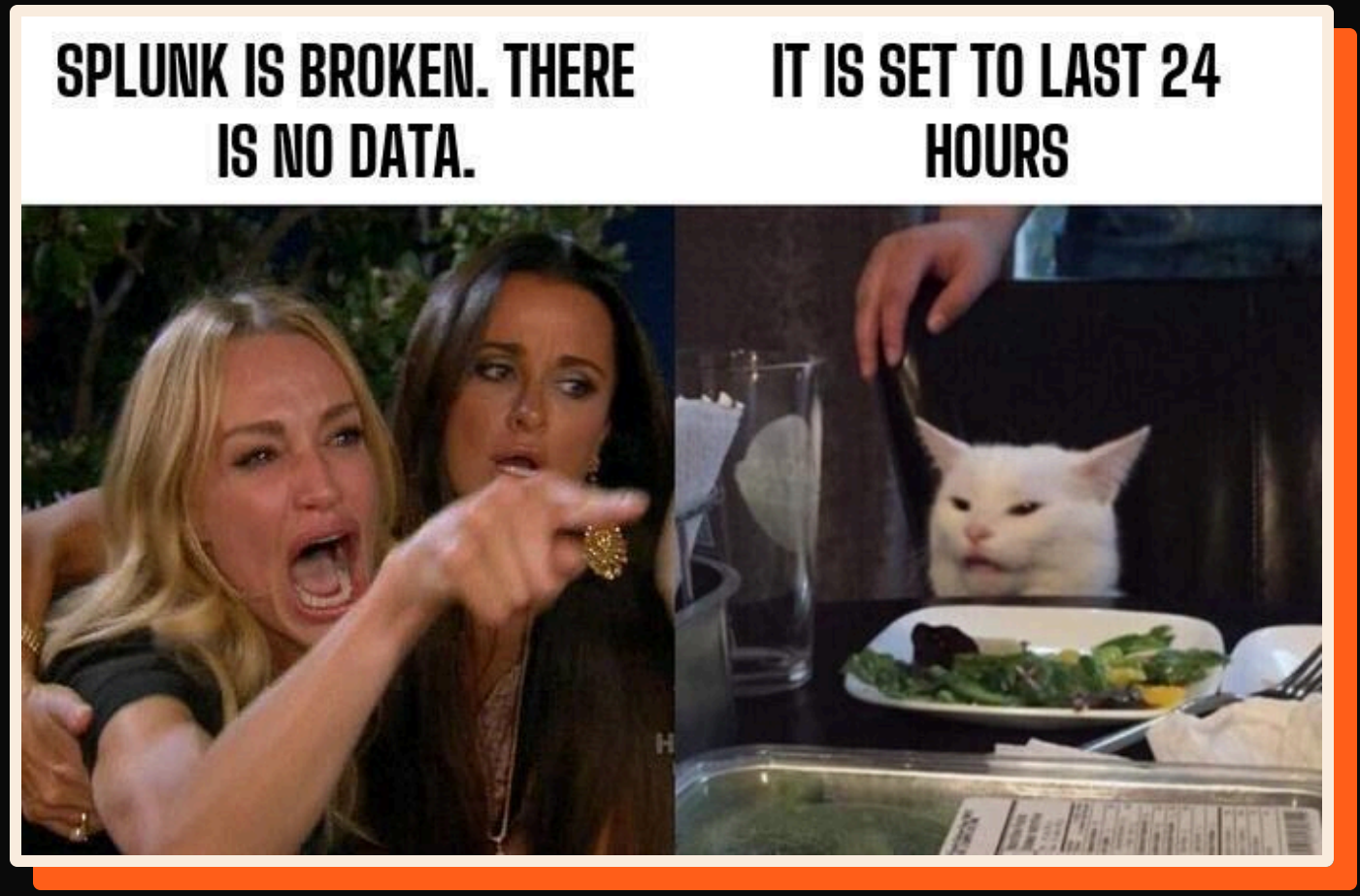
> "CRON GOBLIN uses a post-install script to spawn a shell on developer workstations; evidence is a `node` → `cmd.exe` parent/child in EDR." **If you can write it, you can hunt it.**

WRITE H-0004 DOWN — TWO WAYS, SAME FILE

```
# The flags way (it's in commands/m1-hunt-new-H0004.sh — run the file, don't retype it)
athf hunt new --title "Trojanized build-CLI post-install spawns shell + recon" \
  --technique T1195.002 --tactic execution --tactic initial-access \
  --actor "CRON GOBLIN" --behavior "post-install script spawns a shell that runs recon" \
  --hunter "<you>" --non-interactive

# The no-flags way
cp templates/H-0004.md hunts/   # metadata pre-filled; you fill Check + Keep
athf hunt validate
```

BOTH PRODUCE `HUNTS/H-0004.MD`. PICK THE ONE THAT GETS YOU BACK TO HUNTING FASTER.



SPLUNK ISN'T BROKEN. CHECK THE TIME PICKER.

LAB 1 — 25 MINUTES

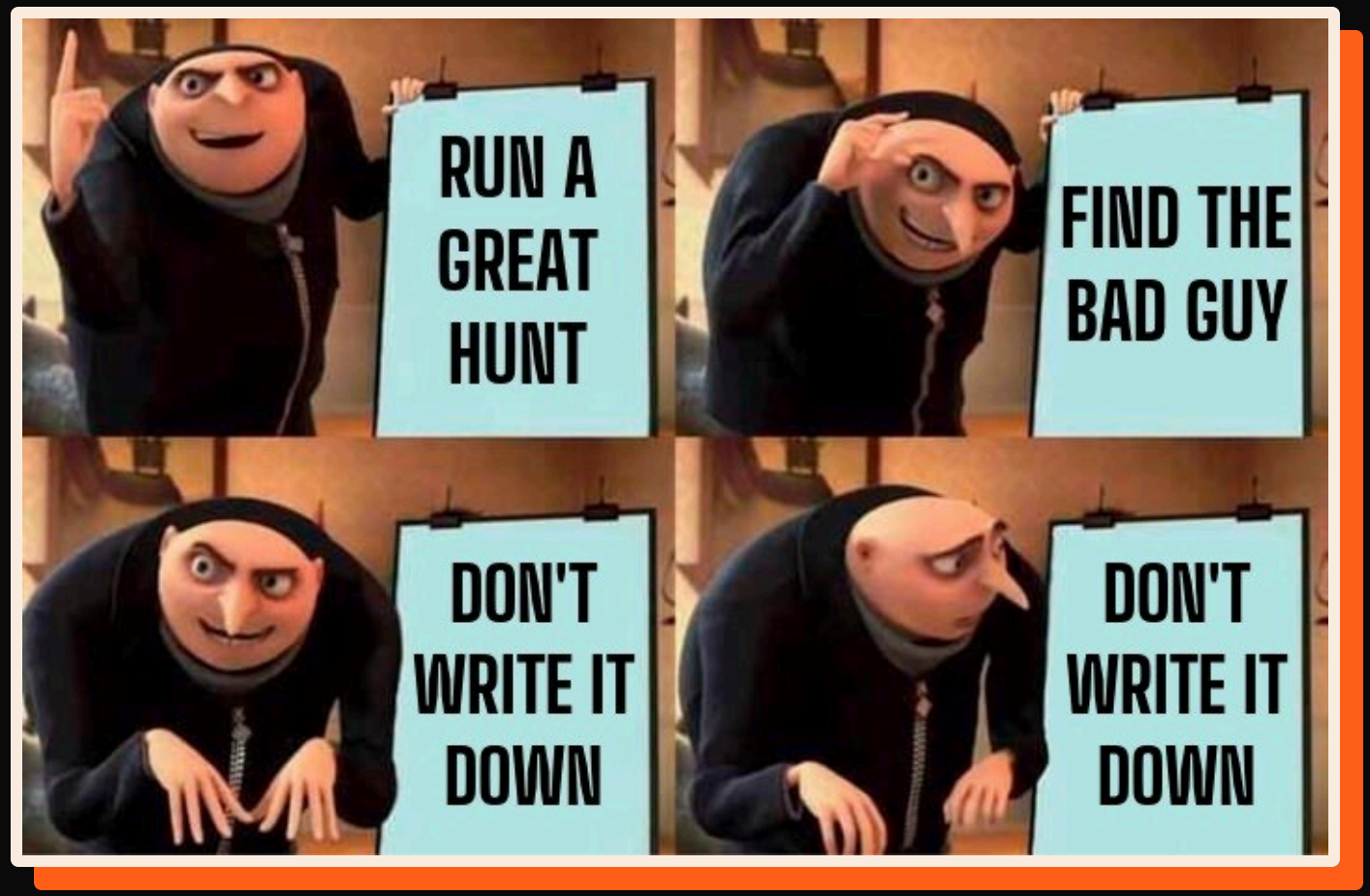
- Hunt Sudo Shipping by hand in Splunk. Workbook Module 1; queries in `queries/m1.md`.
- Write `H-0004` (post-install chain, *confirmed*) and `H-0005` (outbound after install, *needs-follow-up*) as you go — flags or template.
- "Done" = a re-runnable hypothesis, the exact SPL, a verdict, and what you ruled out.
- Time picker: `All time`.

PAUSE THE VIDEO HERE. COME BACK FOR THE DEBRIEF.

DEBRIEF — WHAT YOU SHOULD HAVE FOUND

```
index=thrunt sourcetype="evidenceforge:ecar:json" object=PROCESS action=CREATE
"node.exe" "cmd.exe"
properties.parent_image_path="*\\node.exe" properties.image_path="*\\cmd.exe"
| table _time hostname principal properties.parent_image_path properties.command_line
```

- **A1** `packmule-cli@2.4.1` install → `node postinstall.js` on `WS-BUILD-01`
- **A2** that `node.exe` spawns `cmd.exe` — an installer opening a shell
- **A3** the shell runs `whoami`, `net group`, `ipconfig` within ~2 min
- **H-0004** *confirmed*. **H-0005** (the beacon) *needs-follow-up* — on purpose.



THE HUNT YOU DON'T WRITE DOWN IS THE HUNT YOU RUN AGAIN.

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SEARCH & SIMILARITY

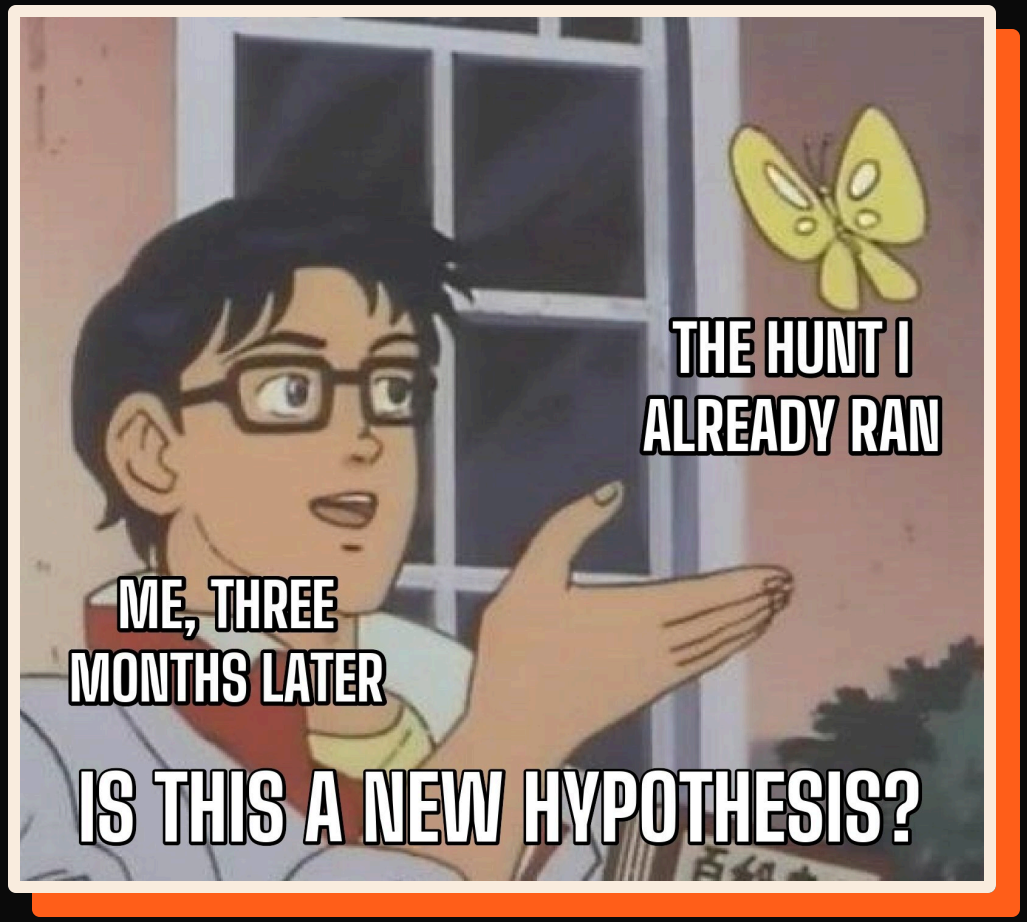
THE HUNT YOU WROTE DOWN STARTS PAYING RENT.

THE RIGHT COMMAND TURNS "NO" INTO "OKAY." SO DOES THE RIGHT CORPUS.

THE QUESTION MODULE 1 COULDN'T ANSWER

- You found WS-BUILD-01. **Was it the only victim?**
- Search the same host, same day, same command — and you miss the second one entirely.
- CRON GOBLIN time-shifted install two: over a day later, different developer's box, inside a legit build window.

> Keyword search only finds more of exactly what you already saw.



WITHOUT MEMORY, EVERY HUNT LOOKS LIKE A NEW ONE.

BEHAVIOR IS THE REUSABLE UNIT

- In H-0004 you wrote a shape: **trusted-dev-process → unexpected shell → recon child.**
- That sentence is host-agnostic, time-agnostic, user-agnostic.
- Indicators rot — the IP, the domain, the hostname all change on the next box.
- Pivot the *shape* across all hosts and all time, and the second victim falls out.

TWO COMMANDS DO THE L2 WORK

COMMAND	QUESTION IT ANSWERS
`athf similar --hunt H-0004`	"WHAT HAVE I HUNTED LIKE THIS BEFORE?"
`athf similar "post-install shell recon"`	FREE-TEXT AGAINST YOUR CORPUS
`athf hunt coverage`	"WHAT HAVE I *NOT* HUNTED?"
`athf hunt coverage --detailed`	TECHNIQUE-TO-HUNT MAPPING

LAB 2 — 20 MINUTES

- Orient: `athf similar` / `coverage` off your Module 1 hunts. Queries in `queries/m2.md`.
- Pivot the H-0004 shape across every host and all time. Find the **`second-wave victim`** (A4).
- Write **`H-0006`** (`templates/H-0006.md` if you'd rather not type flags) and name what found it.
- Time picker: **`All time`**.

PAUSE THE VIDEO HERE. COME BACK FOR THE DEBRIEF.

DEBRIEF — THE HIDDEN VICTIM

```
index=thrunt sourcetype="evidenceforge:ecar:json" object=PROCESS action=CREATE
"cmd.exe" ("whoami.exe" OR "systeminfo.exe" OR "net.exe" OR "ipconfig.exe")
properties.parent_image_path="*\\cmd.exe"
properties.image_path IN ("*\\whoami.exe","*\\systeminfo.exe","*\\net.exe","*\\ipconfig.exe")
| stats min(_time) as first values(properties.image_path) as cmds by hostname principal
```

- Indicator search (`WS-BUILD-01`, `packmule-cli`): **one host**, even on All time.
- The shape, host-agnostic: `WS-DEV-04` **/ carol.grep**, ~25 h later. That's **A4**.
- `sharp@0.33.2` → `node-gyp rebuild` looks similar and is **benign**: it compiles, no recon.
- **H-0006** note: *"keyword search failed; the behavioral pivot off H-0004 found this."*

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AI RESEARCH AGENTS

THE AGENT GENERATES THE HYPOTHESIS. SPLUNK CONFIRMS IT.

RESEARCH IS THE STEP BEFORE THE HUNT

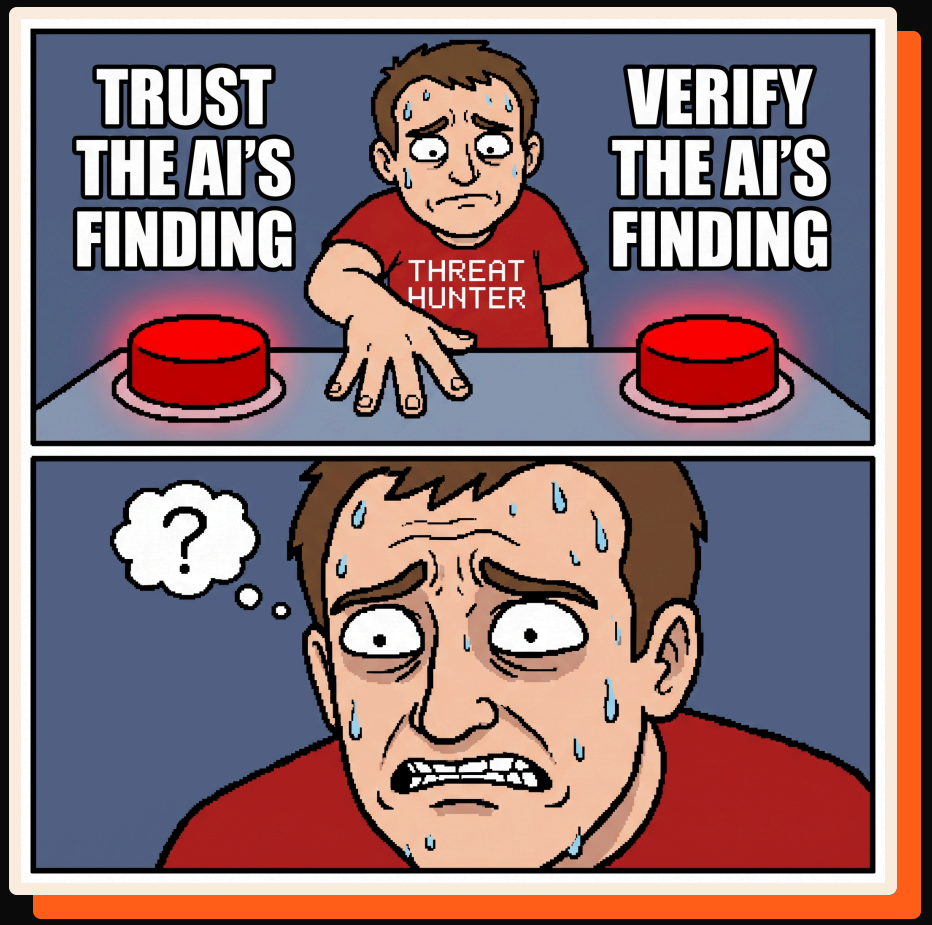
- Everything so far was **hunting**: you had a hypothesis, you tested it.
- **Research** is the step *before* the hypothesis — read intel, understand tradecraft, generate directions.
- Two objects in ATHF: a **research doc** (`athf research new`) proposes; a **hunt** (`athf hunt new`) tests.
- Research feeds hunts. Link them: `athf hunt new --research R-0001`.

WHERE YOU'RE STUCK WITHOUT RESEARCH

- Module 1 left an open thread: H-0005's beacon to `cdn-packmule-telemetry.net`.
- You **observed** it. You couldn't **characterize** it.
- Is 5-minute periodicity meaningful? How would you link this domain to *other* actor infra? Is there a low-and-slow second channel?
- **That's a knowledge gap, not a data gap.** The events are right there.

L3 IS THE FIRST PLACE THE AI CAN BE CONFIDENTLY WRONG

- **Research generates hypotheses; it does not confirm them.** Confirmation lives in Splunk, not the doc.
- **It assumes fields you don't have.** The agent says pivot on JA3 — your Zeek `ssl` has no `ja3` column. Adapt the *idea* to the field you have: the shared hosting block `203.0.113.44/30`.
- **It costs tokens and time.** `--depth basic --no-web-search` in the lab. Research deliberately, not reflexively.



THERE'S ONLY ONE BUTTON. VERIFY. ALWAYS VERIFY.

LAB 3 — 15 MINUTES

```
athf research new --topic "HTTPS C2 beacon detection via TLS JA3 pivoting and shared infrastructure; git-hook persistence in developer environments" \
  --technique T1071.001 --depth basic --no-web-search      # commands/m3-research-new.sh — start it FIRST
```

- While it runs: hunt the git-hook persistence (A5). Then read the doc — every concrete claim is a lead, not a fact.
- Pivot on what it tells you. Confirm in Splunk. Write `H-0007` with `--research R-XXXX`.
- Time picker: `All time`.

PAUSE THE VIDEO HERE. COME BACK FOR THE DEBRIEF.

DEBRIEF — THE SECOND CHANNEL

```
index=thrunt sourcetype="bro:ssl:json" id.resp_h="203.0.113.44/30"
| stats count values(id.resp_h) as ips by server_name
```

- Research said *pivot on JA3*. `ssl.log` has no `ja3`. **The idea survives; the field changes.**
- Same /30, second domain: `assets-packmule-cdn.com` (.45), six callbacks over days. **A8**.
- Decoy `static.docs-cdn.io` sits at .60, outside the /30. Ruled out.
- **A5** `core.hooksPath` → `pre-commit.cmd` → `node loader.js --beacon`. Same shape, aimed at git.

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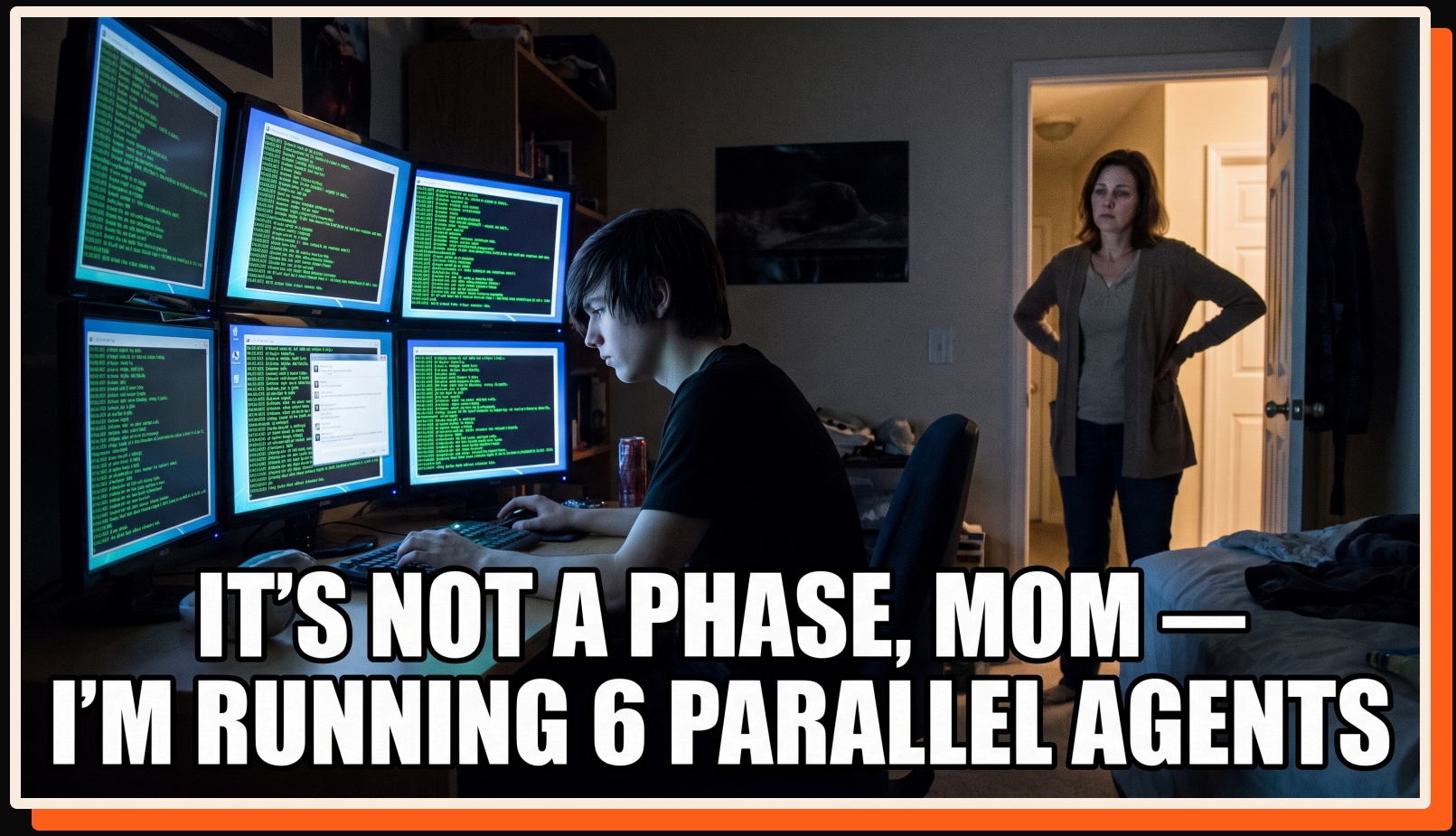
FULL AGENTIC HUNTING VIA MCP

THE AGENT READS YOUR REPO AND NAMES WHAT YOU MISSED.

WHAT CHANGES AT L4

- **L3:** the agent read *the world* — threat intel — and gave you hypotheses.
- **L4:** the agent reads *your work* — your corpus, your coverage — and names *your* blind spots.
- L3 explains DNS tunneling in general. L4 says: **"you have C2 and cred-access covered, nothing on exfil — here's the query."**
- It knows what *you* haven't done because it can see your repo.



AT L4 THIS ISN'T A BIT. PARALLEL AGENTS, ALL READING YOUR REPO.

MCP IN ONE SLIDE — AND THE FIVE TOOLS

TOOL	WHAT IT DOES
`generate_hunt_hypothesis`	PROPOSE A HYPOTHESIS FROM INTEL
`search_hunts_by_text`	FULL-TEXT SEARCH ACROSS YOUR CORPUS
`find_similar_hunts`	THE MCP FORM OF `ATHF SIMILAR`
`list_coverage_gaps`	**THE STAR:** WHICH PHASES HAVE NO HUNT YET
`propose_next_query`	TURN THE BIGGEST GAP INTO A CONCRETE QUERY

MCP HANDS THE AGENT CALLABLE TOOLS. THESE FIVE READ YOUR REPO, NOT A SUMMARY OF IT.

WHY COVERAGE-GAP DETECTION IS THE WHOLE POINT

- Modules 1–3 found what you *went looking for*.
- The two hardest artifacts — cred-access (A9) and DNS-tunnel exfil (A10) — share one property: **you don't know you haven't found them.**
- No open thread points at them. At L0–L2 you simply *stop*.
- `list_coverage_gaps` reads your hunts, maps kill-chain phases, reports: **cred-access ✕, exfil ✕.**

YOU DRIVE IT IN PLAIN ENGLISH

- You will **not** type `list_coverage_gaps`.
- You ask what you'd ask a senior hunting partner: **"What have I hunted?"** / **"What should I hunt next?"**
- The agent picks the tool. Your corpus becomes something you *talk to*.
- Tool names stay as a fallback — offline, or when a question misfires.

LAB 4 — 15 MINUTES

- Ask your repo what you've hunted. Then ask what to hunt next.
- Chase the two Tier-3 gaps it names — cred-access and DNS-tunnel exfil. Queries in `queries/m4.md`.
- Produce `H-0008`. Handshake broken? CLI fallbacks are in the workbook — don't burn the module on config.
- Time picker: `All time`.

PAUSE THE VIDEO HERE. COME BACK FOR THE DEBRIEF.

DEBRIEF — THE GAPS NOBODY POINTED AT

```
index=thrunt sourcetype="bro:dns:json" qtype_name=TXT
"exfil-packmule.net" query="*.exfil-packmule.net"
| eval label=mvindex(split(query,"."),0) | eval len=len(label)
| stats count avg(len) as avg_label_len by id.orig_h
```

- `list_coverage_gaps` over four hunts → **cred-access ✕, exfil ✕.** Your repo pointed there; your data didn't.
- **A9** the beacon reads `.env`, `.aws\credentials`, `.netrc` — theft hiding in normal file reads.
- **A10** long, high-entropy TXT subdomains to `exfil-packmule.net` — exfil hiding in DNS noise.
- **H-0008** `--threat-context`: *"found via list_coverage_gaps."*

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SAME ACTOR, NEW TTP

SAME ACTOR. NEW BATTLEFIELD. 15 MINUTES. GO.

CRON GOBLIN. THE ACTOR YOU'VE HUNTED ALL SESSION JUST CHANGED THE GAME.

THE PIVOT

> "New intel on CRON GOBLIN — the same actor behind the trojanized `packmule-cli` — shows they've pivoted. No more waiting for developers to install a bad package. They've compromised a maintainer's access to a trusted internal repo and are injecting malicious build steps straight into your CI/CD pipeline. The runner does the work now. Same actor. Same infrastructure. New technique. The new campaign is already in your Splunk. You have 15 minutes. Go."

WHY THE PREPARED HUNTER WINS

- **Documented Campaign 1:** a repo full of hunts on C2, cred-access, exfil — the exact behaviors CRON GOBLIN reuses.
- `athf similar` + MCP links the new artifacts to old hunts in seconds. **"Same actor, new host"** falls out for free.
- **Worked ad-hoc (L0):** you'd have nothing to reuse. A blank search bar against unfamiliar CI telemetry.

> Watch the gap between those two hunters. That gap is the product.

THE UN-HUNTED ATTACKER'S WHOLE ADVANTAGE IS TIME YOU DIDN'T SPEND LOOKING.

▮▮ LAB 5 — 15 MINUTES. THE CLOCK STARTS WHEN YOU PAUSE.

- Fast route: the `curl | sh` build step nobody committed → ask your repo *"does this share infra with anything I've hunted?"* → **`H-0009`**.
- Cite the earlier hunt it reused. That citation is the whole story in one line.
- Ask your repo before you open the answer key.
- Time picker: **`All time`**.

PAUSE THE VIDEO HERE. DON'T SCRUB AHEAD — THE NEXT SLIDE IS THE ANSWER.

DEBRIEF — SAME ACTOR, PROVEN

```
index=thrunt sourcetype="apache:access:json" server="cdn-packmule-telemetry.net"
| stats count values(user) as users by client
```

- **M5-A1** `CI-RUNNER-03` runs `curl -fsSL … | sh` — a build step nobody committed.
- **M5-A3** ★ one proxy query: a Campaign-1 workstation **and** the runner, **same C2 domain**. Same actor.
- Stretch **M5-A2** IMDS cred theft (your H-0008 shape). **M5-A4** `curl -T` to an attacker bucket; `sudo-releases` is the legit deploy.

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WHAT YOU BUILT, AND WHAT'S NEXT

SIXTY MINUTES OF VIDEO, NINETY OF HUNTING. SIX HUNTS. ALL YOURS.

THE ARC YOU JUST WALKED

LEVEL	WHAT YOU DID	TOOL
L0	HUNTED IN A SEARCH BAR.	—
L1	WROTE THE HUNT DOWN.	`athf hunt new`
L2	BEHAVIOR FOUND A HIDDEN VICTIM.	`athf similar`
L3	AN AGENT NAMED WHAT TO HUNT.	`athf research new`
L4	AN AGENT NAMED YOUR BLIND SPOT.	MCP

EACH LEVEL ONLY WORKED BECAUSE THE ONE BELOW IT WROTE SOMETHING DOWN.

THE MOMENT THAT MATTERED

- **Module 5**: Same actor, new battlefield, 15 minutes.
- Documented hunters linked the CI beacon to Campaign-1 infra in seconds — *"same actor, new host."*
- The ad-hoc hunters started **cold**.

THAT GAP IS THE WHOLE THESIS: MEMORY IS THE EDGE. NOT A BIGGER MODEL. THE FACT THAT YOU WROTE IT DOWN.

BEHAVIOR OVER INDICATORS

- The binary was signed. The hash was clean. There was no CVE.
- Every artifact you found today was **behavioral** → a shape that transfers across hosts, time, and infrastructure.
- Indicators rot. Behavior persists. It is the durable signal.

THAT'S NOT A SLOGAN. YOU JUST SPENT NINETY MINUTES PROVING IT.

BURY THE INDICATOR. KEEP THE BEHAVIOR.

"WHAT HAPPENS AT 500 HUNTS?"

- Someone asks this every time. Good question. Short answer: **the agent never reads the whole corpus.**
- `athf similar` **ranks**, it doesn't read — the corpus is vectorized, the agent sees the top matches.
- `list_coverage_gaps` is an **aggregate** over hunt metadata, not a pass over every record.
- Token caps are per call. A big corpus makes the ranking *better*, not the context bigger.
- And hunts are files in a repo. Git scales. Prune, tag, archive like any other codebase.

IT'S ALL YOURS

- **ATHF is open source, on PyPI.** Vendor-neutral by design.
- Anthropic, OpenAI, Bedrock, Ollama — whichever key you brought, it ran. Vendor-neutral by design.
- Nothing you learned is locked behind a login.

TAKE THE REPO. TAKE THE WORKFLOW. RUN IT AGAINST YOUR OWN TELEMETRY MONDAY.

TAKE THE REPO. TAKE THE WORKFLOW. RELISH THE HUNT.

THE SIBLING QUESTION

- You hunted these behaviors *once*.
- The obvious next question: how do you catch them *every* time, automatically, with no human in the loop?
- **Hunting finds the unknown. Detection operationalizes the known.**

THAT HAND-OFF — HUNT → DURABLE DETECTION — IS WHERE MOST TEAMS LOSE THEIR WORK.

ADEF — THE SIBLING FRAMEWORK

- **ADEF — the Agentic Detection Engineering Framework.** ATHF's sibling.
- ATHF asks *"what happened that we didn't know about?"*
- ADEF asks *"how do our findings become detections that never miss it again?"*
- Same philosophy — behavior-first, memory-first, AI-augmented, open — pointed at the detection lifecycle.

TAKE THESE WITH YOU

- `ATHF` — `pip install agentic-threat-hunting-framework` · github.com/Nebulock-Inc
- `ADEF` — the sibling, same org: your hunts become durable detections
- `Workshop materials` — github.com/Nebulock-Inc/agentic-hunting-workshop-deathcon-2026
- `Live Q&A at DEATHcon` — ask in the workshop Discord channel. Bring your H-0009 and what stumped you.
- `The survey` — 90 seconds. It shaped this version.

THANK YOU FOR HUNTING WITH US.

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ONE LAST THING

Poked at the weird stuff?
The hot-dog service account, the Pokémon on the dev VLAN, the `hunter2` in a `.env`? This one's for you.

THE GLIZZARD SALUTES THE CURIOUS. STAY CURIOUS. HUNT WEIRD.